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Mass Spectrometry of Inductively Coupled Plasmas

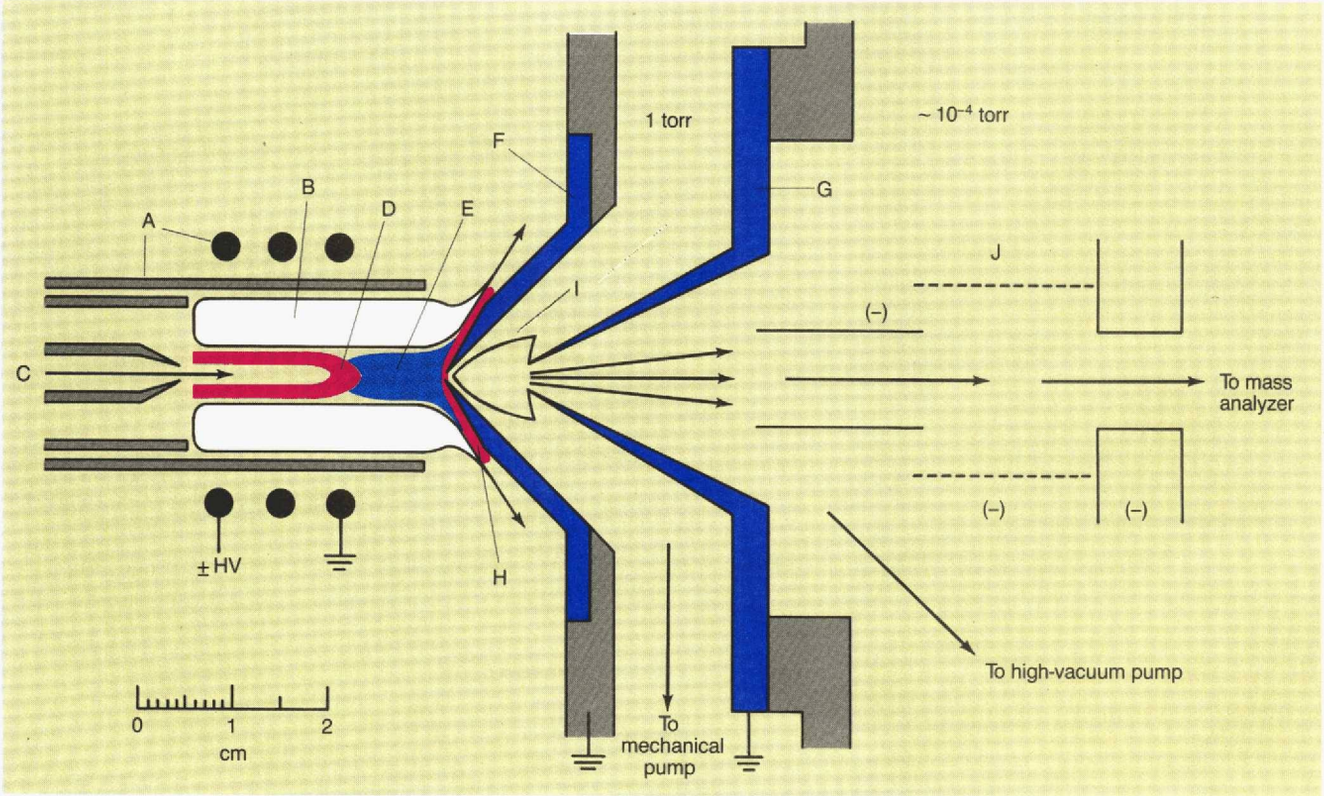
The field of elemental analysis seems to experience periodic upheavals as new techniques and instrumentation are devised to carry out this oldest of analytical tasks. Gravimetry was superseded by colorimetry, then came atomic absorption spectrometry, which was followed by inductively coupled plasma–atomic emission spectrometry (ICP–AES). Each of these techniques has been used widely by the analytical community during the time in which it proved either complementary or superior to other ways of solving important analytical problems.

Recent experimental work has demonstrated the potential capabilities of yet another challenger in this arena, namely mass spectrometry (MS) with an ICP as ion source. This new concatenation of acronyms (ICP–MS) grew from expertise available within several independent lines of investigation:

- development of the ICP as a superior excitation source for analysis of solutions by AES (*1–3*),
- development of techniques for sampling flames and plasmas for fundamental studies of their structure and

Figure 1. ICP and sampling interface. A solution of Y is injected into the plasma with resulting red emission from excited Y or YO (*13*) and blue emission from Y^+

(A) torch and load coil (HV = high voltage), (B) induction region of ICP, (C) solution aerosol being injected into axial channel, (D) initial radiation zone, (E) normal analytical zone, (F) nickel cone with sampling orifice in tip, (G) skimmer cone, (H) boundary layer of ICP gas deflected outside sampling orifice, (I) expanding jet of gas sampled from ICP, and (J) ion lens elements



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properties by MS (*4*), and
- the original demonstration by Gray that useful mass spectra of elemental constituents in solutions could be obtained from a dc plasma (*5*, *6*).

The first analytical mass spectra from an ICP were obtained by the author in 1978 (*7*). Sciex, Inc., introduced the first commercial instrument for ICP–MS at the 1983 Pittsburgh Conference, which catalyzed an explosive growth of interest in the technique. At present there are two commercial manufacturers of ICP–MS instrumentation (the other is VG Isotopes). Several other instrument companies are conducting ICP–MS research, and approximately 50 such instruments are in operation. The purpose of this article is to survey the basic principles and analytical capabilities of ICP–MS. More comprehensive data and information are available in the recent literature (*8–12*).

The ICP as an ion source

From an analytical perspective, it has long been realized that emission lines from monatomic, singly charged, positive ions are generally the most prominent or useful for AES with the ICP (*13*). Consider the chemical status of analyte species in the normal analytical zone (*14*), which is depicted in the blue region (E) in Figure 1. At this position the gas kinetic temperature is approximately 5000 K (*3*). Furthermore, the injected analyte species have spent approximately 2 ms in the hot plasma and thus are efficiently atomized. The efficiency of ionization for the resulting analyte atoms may be estimated from the Saha equation (*15–17*); the results of such calculations are shown in Figure 2. From this compilation it is apparent that formation of singly charged ions is very efficient in the ICP. Some 54 elements are expected to be ionized with an efficiency of 90% or more. Even metalloid or nonmetallic elements such as P and As, for which ionic emission lines are either absent or not prominent, are ionized with reasonable efficiency. In a pure Ar ICP, Ar^+ is expected to be the major ion present despite its low ionization efficiency. H^+, O^+, and other ions containing H and O would also contribute significantly to the total ion yield from an ICP containing aqueous aerosol.

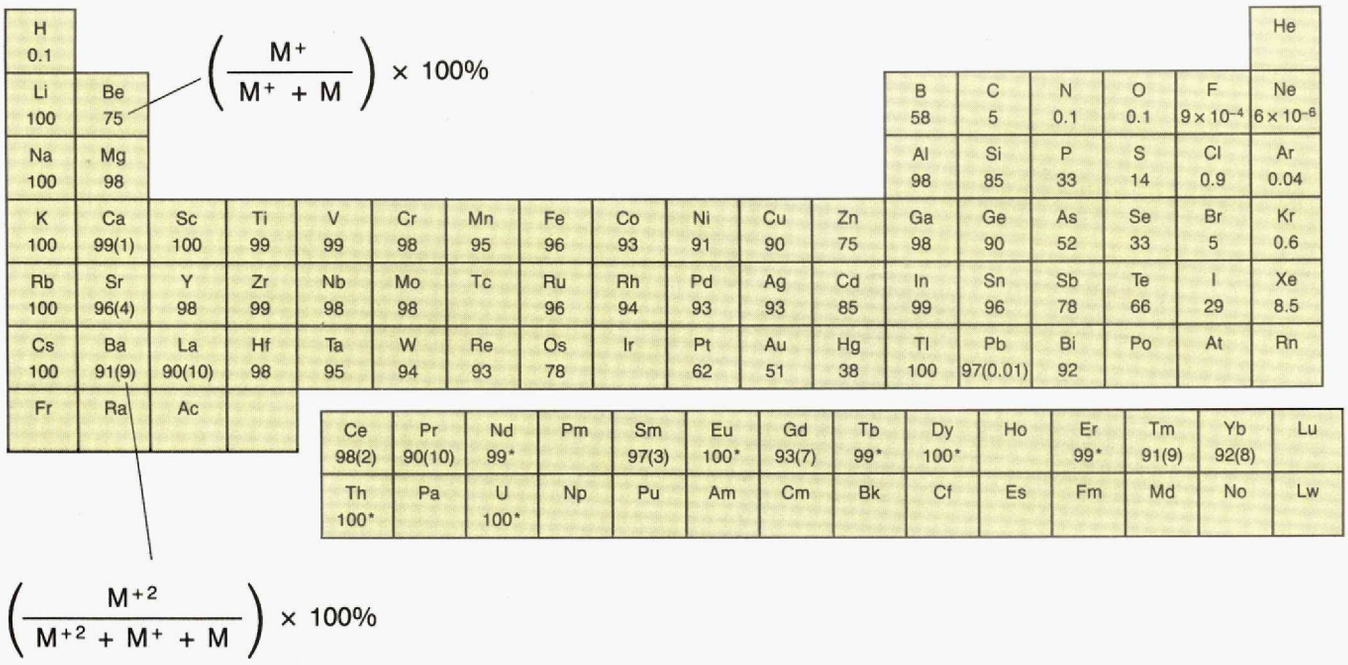
The extent to which multiply charged ions are present in the ICP can also be estimated from the Saha equation. Such calculations are shown in parentheses in Figure 2 for those elements with the lowest second ionization energies and for which the necessary partition functions are available (*18*). The highest level of doubly charged ion formation expected for any element is only approximately 10% of the total analyte ion density. These levels of M^{2+} are expected only for Ba and several rare earths. Most elements have second ionization energies comparable to or greater than that for Pb (15 eV, see Figure 2) so that doubly charged ions should be negligible. Ions of higher charge require even more energy and are also negligible. Thus the ICP is a very efficient, but not extremely energetic, ion source in that it produces a high yield of M^+ and little else. There are some elements for which oxide ion peaks (MO^+) are also observed as described below. In addition to being an efficient source of ions, the ICP exhibits greater immunity to chemical-type interference effects than any other spectrochemical discharge (*19*, *20*).

Extraction and detection of ions from the ICP

The preceding discussion indicates the ICP to be a potentially effective ion source. The rub, of course, lies in the fact that it operates at atmospheric pressure, whereas MS requires an operating pressure $\lesssim 10^{-5}$ torr for resolution and detection of ions. The mass spectrometrist's classic solution to such a pressure discrepancy between gas source and analyzer is to interpose an interface between the two. The development and improvement of the ion sampling interface for ICP–MS have been crucial aspects of the success of the technique.

Figure 1 depicts the present arrangement used at the Ames Laboratory and in the commercial instruments for extracting ions from the ICP. The plasma flows around the tip of a water-cooled metal cone called the sampler. This cone has a circular orifice of 0.5–1.0-mm diameter drilled into its tip. Gas from the ICP enters this orifice from a cross-sectional area of diameter approximately three times the orifice diameter, so that a 1.5–3-mm-wide section of the axial channel is sampled. This represents a large fraction of the total gas in the axial channel. Most of the gas flow is evacuated by a mechanical pump that maintains a pressure on the order of 1 torr.

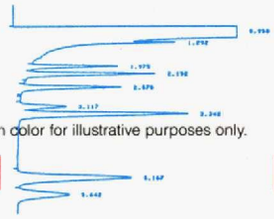
The central orifice of a conical



H 0.1																	He
Li 100	Be 75											B 58	C 5	N 0.1	O 0.1	F 9×10^{-4}	Ne 6×10^{-6}
Na 100	Mg 98											Al 98	Si 85	P 33	S 14	Cl 0.9	Ar 0.04
K 100	Ca 99(1)	Sc 100	Ti 99	V 99	Cr 98	Mn 95	Fe 96	Co 93	Ni 91	Cu 90	Zn 75	Ga 98	Ge 90	As 52	Se 33	Br 5	Kr 0.6
Rb 100	Sr 96(4)	Y 98	Zr 99	Nb 98	Mo 98	Tc	Ru 96	Rh 94	Pd 93	Ag 93	Cd 85	In 99	Sn 96	Sb 78	Te 66	I 29	Xe 8.5
Cs 100	Ba 91(9)	La 90(10)	Hf 98	Ta 95	W 94	Re 93	Os 78	Ir	Pt 62	Au 51	Hg 38	Tl 100	Pb 97(0.01)	Bi 92	Po	At	Rn
Fr	Ra	Ac															

Ce 98(2)	Pr 90(10)	Nd 99*	Pm	Sm 97(3)	Eu 100*	Gd 93(7)	Tb 99*	Dy 100*	Ho	Er 99*	Tm 91(9)	Yb 92(8)	Lu
Th 100*	Pa	U 100*	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lw

Figure 2. Calculated values for degree of ionization for M^+ at T_{ion} = 7500 K, $n_e = 1 \times 10^{15}$ cm^{-3}
Figures in parentheses represent percentages of M^{2+} formed. Elements marked by asterisks yield significant M^{2+}, but partition functions were not available



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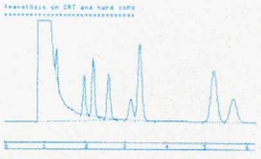
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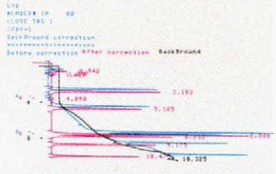
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skimmer is located behind the sampler at an appropriate position to transmit as much of the sampled beam as possible into a second vacuum chamber (*21*, *22*). The pressure here is low enough and the mean free path long enough for ion lenses to collect, focus, and transmit the ions to the mass analyzer. With today's instruments, there is no deliberate source of ionization inside the vacuum system. To be observed by the MS, an ion must be present as such in the ICP and survive the extraction process or be formed by chemical reactions during the extraction process.

Two devices for ICP–MS are depicted in Figure 3. The instruments constructed to date employ a quadrupole mass analyzer (*23*, *24*) and yield essentially unit mass resolution over a mass range up to at least $m/z = 300$. The mass analyzer can be scanned very rapidly ($\leq$20 ms for one sweep from $m/z = 1$ to 200) or the mass transmitted can be switched repetitively from peak to peak with only a few μs required between each jump. However, it is important to emphasize that at any instant only ions of one given m/z value have a stable path through the mass analyzer to the detector. Thus ICP–MS with a quadrupole mass analyzer is not truly a simultaneous multielement measurement in that signals from different elements cannot be measured at the same time. ICP–MS is better described as a sequential multielement measurement in which very little time is lost switching from one element to the next. Signal averaging techniques can be readily employed for prolonged, repetitive data acquisition in either the continuous scanning or "peak hopping" (i.e., multiple ion monitoring) mode.

Ions of the selected m/z value leave the mass analyzer and are deflected into an electron multiplier for detection. Generally, a Channeltron multiplier (*25*) is used in a pulse-counting mode to discriminate against rf background from the mass analyzer and ICP, although analog ion current measurements are also possible. The detector is also sensitive to photons radiated from the ICP through the sampler and skimmer (*26*). It is therefore desirable to block the line of sight through the ion lens with a solid metal baffle (Figure 3b) and/or to offset the multiplier from the system centerline (Figure 3a) to minimize the photon background.

The use of a large sampling orifice for ICP–MS has numerous desirable aspects from a practical viewpoint. The larger the sampling orifice, the greater the flow rate of ions into the vacuum system. The larger the orifice, the more resistant it is to plugging by solids condensed from the injected so-

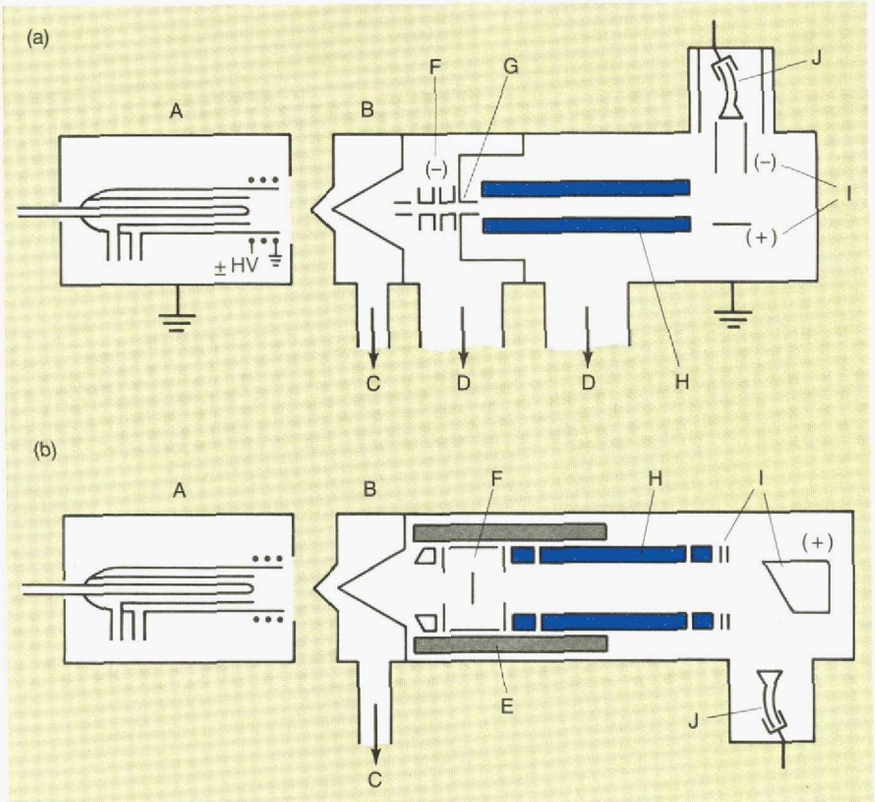


Figure 3. Schematic diagrams of ICP–MS instruments constructed (a) in Ames and (b) by Sciex

(A) ICP, torch, and shielding box, (B) ion extraction interface with sampler and skimmer cones, (C) line to mechanical vacuum pump (~3 L s^{-1}), (D) oil diffusion pumps (1600 L s^{-1}) (3a only), (E) cryogenic pumping surfaces (3b only), (F) ion lens elements (central photon baffle shown in 3b), (G) differential pumping orifice (3a only), (H) quadrupole mass analyzer (rf-only rods are shown at the entrance and exit of the mass analyzer in Figure 3b), (I) ion optical elements for deflecting ions into detector, and (J) offset Channeltron electron multiplier. The instruments used by Gray and Date and those produced by VG Isotopes use a vacuum system similar to that shown in Figure 3a with a photon baffle in the ion lens

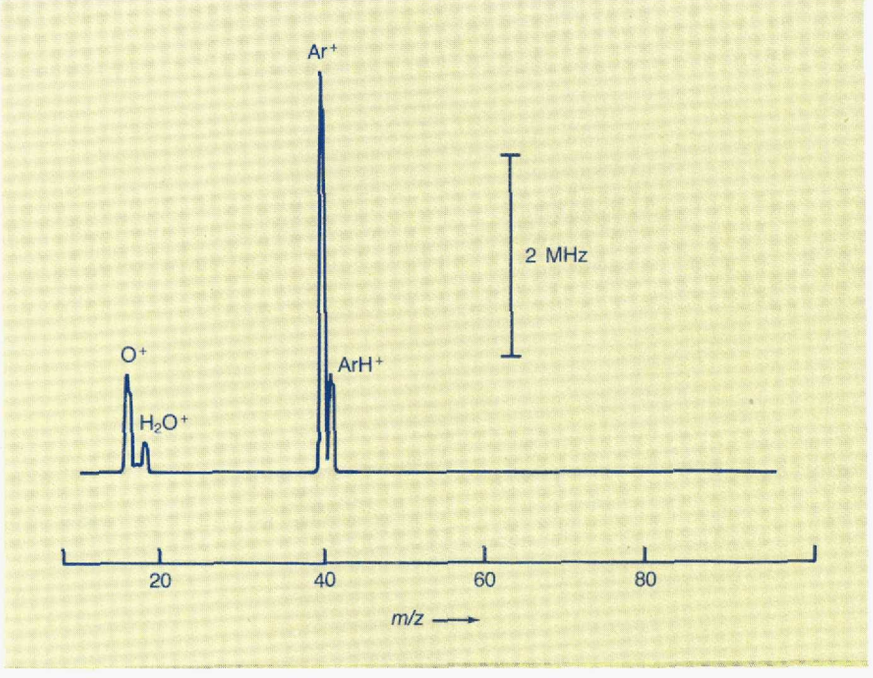


Figure 4. Mass spectrum of ICP showing major background ions detected during nebulization of deionized H_2O

Ion detection here and in Figure 6 was in analog current mode to prevent signal loss when detecting major ions. Vertical scale shows output signal after current-to-frequency conversion. Reproduced from Reference 31

Table I. Additional background ions not evident in Figure 4

Ion[a]	m/z values[b]	Overlap with (% natural abundance)	
O_2^+	32	$^{32}S^+$	(95.0)
	34	$^{34}S^+$	(4.2)
	36	—	
N_2^+	28[c]	$^{28}Si^+$	(92.2)
Ar_2^+	80	$^{80}Se^+$	(49.8)
	76	$^{76}Se^+$	(9.0)
	78	$^{78}Se^+$	(23.5)
	72	$^{72}Ge^+$	(27.4)
	74	$^{74}Ge^+$	(36.7)
		$^{74}Se^+$	(0.9)
ArO^+	56	$^{56}Fe^+$	(91.7)
	52	$^{52}Cr^+$	(83.8)
	58	$^{58}Ni^+$	(67.8)
	54	$^{54}Cr^+$	(2.3)
		$^{54}Fe^+$	(5.8)
NO^+	30[c]	$^{30}Si^+$	(3.1)

[a] Listed in decreasing order of intensity.
[b] Isotopic species also listed in decreasing order of abundance.
[c] Contributions from Si^+ also possible.

lution. The gas flow through a large orifice is sufficient to draw gas directly from the ICP with minimal cooling of the sampled gas outside the vacuum chamber. This phenomenon is depicted in Figure 1 by the constriction of the flow streamlines directly into the sampling orifice. Of course, it is necessary to construct and cool the sampler in such a way that the tip does not melt, erode, or plug for a reasonable time, i.e., at least one day of operation.

It is also desirable that ions from the orifice material do not appear in the mass spectrum so that the substance(s) used to make the sampler can be determined analytically. Nickel has been found to be a useful metal for fabricating the sampling orifice because of its reasonably good thermal conductivity, stability, and machinability.

The first attempts to use a large sampling orifice for ICP–MS were

Table II. Some background ions observed from mineral acid solvents (*9*)

Acid	Ions[a]	m/z	Overlap with (% natural abundance)	
HNO_3	N^+	14	—	
	ArN^+	54	$^{54}Fe^+$	(5.8)
			$^{54}Cr^+$	(2.3)
$HCl,HClO_4$	Cl^+	35,37	—	
	ClO^+	51	$^{51}V^+$	(99.7)
		53	$^{53}Cr^+$	(9.6)
	$ArCl^+$	75	$^{75}As^+$	(100)[b]
		77	$^{77}Se^+$	(7.6)
	HCl^+	36,38	—	
H_2SO_4	S^+	32,33,34	—	
	SO^+	48	$^{48}Ti^+$	(74.0)
		49	$^{49}Ti^+$	(5.5)
		50	$^{50}Ti^+$	(5.2)
			$^{50}Cr^+$	(4.3)
			$^{50}V^+$	(0.3)
	SO_2^+	64	$^{64}Zn^+$	(48.9)
			$^{64}Ni^+$	(1.2)
		65	$^{65}Cu^+$	(30.9)
		66	$^{66}Zn^+$	(27.8)

[a] Listed in decreasing order of abundance for each acid.
[b] Only isotope.

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hampered by an electrical discharge that occurred from the bulk plasma through the gas flow streaming into the orifice to the sampler. This discharge, originally termed the "pinch effect" (*27, 28*), caused melting or erosion of the orifice, excessive multiply charged ions, high ion kinetic energies (*29, 30*), formation of ions from the orifice material, broadened and/or split peaks, and a high photon background. Judicious grounding and shielding of the load coil (e.g., Figure 1) reduce the intensity of the discharge to the point where these deleterious effects should no longer be observed. The author feels that an attenuated discharge does persist in his homemade instruments because of (a) the dependence of ion kinetic energy on plasma operating parameters (*29, 30*) and (b) the visible (pink, orange, or purple) glow observed inside the vacuum system. Douglas and French have described a unique way to ground the load coil that apparently eliminates the discharge completely (*30*).

Analytical figures of merit

One reason Ar is used as support gas for ICP–AES is its transparency and lack of emission lines in the wavelength region 200–400 nm, where many of the most useful analyte lines occur (*13*). The background mass spectrum of the ICP shows a similar simplicity, as indicated in Figure 4 and Table I. Some additional ions commonly detected during nebulization of various aqueous acids are listed in Table II (*10*).

From this information the following general statements about the background spectra are possible. First, the major ions overlap directly with only a few analyte peaks of interest; the worst cases are with $^{40}Ca^+$ and $^{32}S^+$, the major isotopes of Ca and S. The minor ions, such as ArO^+ and Ar_2^+, must be anticipated and compensated for during determination of trace levels of Fe and Se, respectively, although there is a free Se isotope at $m/z = 82$ (9.2% abundant). A similar situation holds for the acid-derived ions such as ArN^+, ClO^+, $ArCl^+$, SO^+, and SO_2^+. The intensities of these ions should be constant between standards and samples provided the same solvent composition is used. No appreciable background ions are observed above $m/z = 80$.

Many (if not all) of the minor molecular ions are not present as such in the ICP but are probably formed by clustering reactions during the ion extraction process. There is considerable interest in attenuating these ions by further refining the ion sampling interface to prevent their formation, or by dissociating them in a collision chamber between the interface and the detector. Another area of research interest is simplification of the background spectra from organic or mixed aqueous–organic solvents, which yield a greater number of more intense peaks than are observed from aqueous solutions.

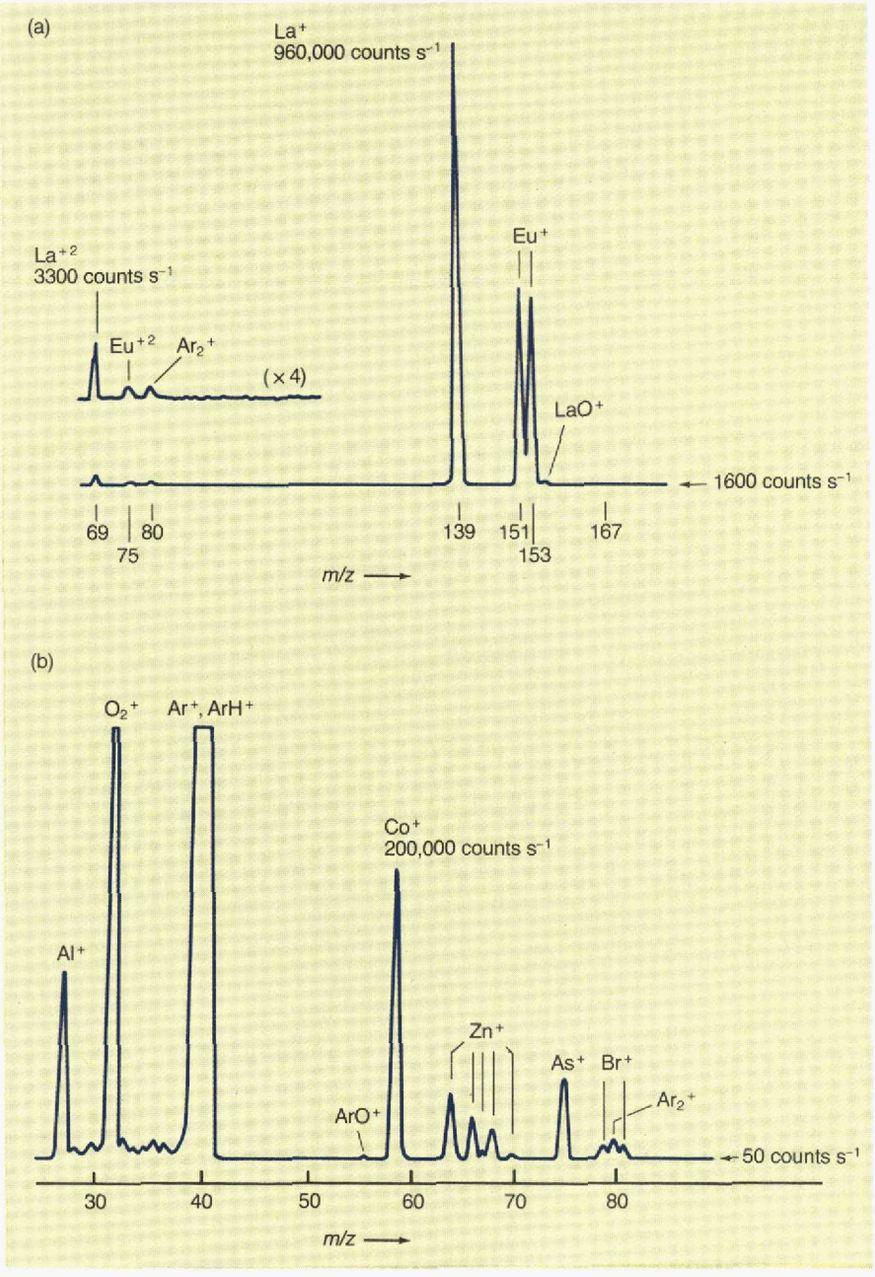


Figure 5. Some representative analyte spectra obtained using pulse-counting detection

(a) Spectrum of 1 mg L^{-1} La and equimolar Eu taken with instrument shown in Figure 3a. Note that no EuO^+ is observed at $m/z = 167, 169$. (b) Spectrum of 1 mg L^{-1} Co and equimolar Al, Zn, As, and Br taken with the instrument shown in Figure 3b. The same nebulizer (*32, 33*) was used with both instruments

Mass spectra of solutions of several representative elements are shown in Figure 5. The spectra were obtained under plasma operating conditions generally found useful for multielement analysis and with the mass filter operating under relatively low resolution to minimize mass discrimination. Lanthanum (Figure 5a) was chosen as an element with one of the most complicated mass spectra because of its low second ionization energy (11.06 eV) and its tendency to form refractory oxide species. Relative to La^+ the observed levels of La^{2+} and LaO^+ are 3% and 0.8%, respectively. These levels are representative of those expected for rare earths, alkaline earths, Th, and U. Most elements yield no significant M^{2+} or MO^+ as shown by the other spectra in Figure 5. It is possible to change ICP parameters to deliberately make more MO^+ (or M^{2+} in some instruments); the aerosol gas flow rate is the parameter to which the spectrum is most sensitive (*10, 29, 34*).

The observed mass spectra are sim-

ple enough for unit mass resolution to be sufficient for most purposes. There are cases where the presence of even low levels of MO^+ or M^{2+} from the major sample constituents must be anticipated. For example, consider determination of trace Cd (m/z = 106, 108, 110–114, 116 for Cd^+) in the presence of a large excess of Mo. As Date et al. point out (*35*), the residual MoO^+ obscures all the Cd isotopes except for the first one, which is only 1.2% abundant naturally. Oxide ions and doubly charged metal ions from the rare earth elements would cause similar problems if their presence were not suspected. Fortunately, it is easy to check for such problems by obtaining m/z scans of actual samples to be analyzed. Quality control analysis and a healthy distrust of the data generated are as necessary in ICP–MS as in other techniques.

If the isotopic peaks for each element in Figure 5b are summed, the order of sensitivity (i.e., count rate per unit concentration) is Co > Zn > As > Br. This sequence follows from the degrees of ionization shown in Figure 2 and the relative ionization energies of these elements, i.e., Co (7.86 eV) < Zn (9.39) < As (9.81) < Br (11.81). The sensitivity for Zn and As relative to Co^+ is a bit less than would be expected from Figure 2, but the point is that even elements with high ionization energies like Br can be detected with useful sensitivity.

The observation of less Al^+ than expected relative to Co^+ illustrates that the transmission can differ significantly for ions of widely different mass depending on the operating conditions chosen for the ion optics and mass filter. It is also interesting to note from Figure 5 that although the analyte ion count rate and the background are higher for the instrument without the photon baffle (Figure 3a), the instrumental configuration shown in Figure 3b yields better detection limits. There may yet be considerable room for improvement in ways of transmitting and detecting ions at the expense of photons.

Commercial instruments for ICP–MS are capable of yielding analyte count rates of 10^5–10^6 counts s^{-1} per mg L^{-1} on top of a background that for most elements is 50 counts s^{-1} or less. If the standard deviation of the background is 10 counts s^{-1}, the detection limit is the solution count rate necessary to yield a net signal of 30 counts s^{-1} or so. This count rate corresponds to detection limits of 0.03–0.3 $\mu g\ L^{-1}$. This range represents the detection limits obtainable under multielement operating conditions. For many elements, even better detection limits can be obtained if operating conditions for the plasma, ion optics, and mass analyzer are optimized exclusively for one element or a few elements of similar mass. Naturally, detection limits measured on a mass-per-volume basis are best for lighter elements with one dominant isotope and ionization energies below 8 eV.

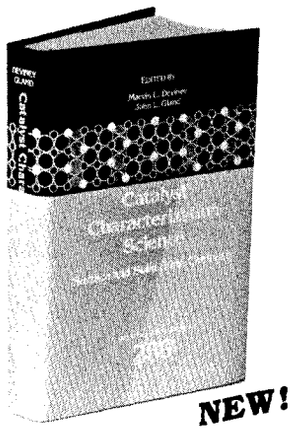
The detection limits obtainable by ICP–MS are considerably better than typical values reported for ICP–AES (0.1–1 $\mu g\ L^{-1}$) with the same type of pneumatic nebulizer (*1–3*). Much of the interest in ICP–MS originated from users of ICP–AES who found the latter approach to have inadequate detection limits or too many cases of overlapping spectral lines for their applications. As with most analytical techniques, true quantitative analysis requires analyte concentrations somewhat above the detection limit. For example, the standard deviation of an analyte peak at 30 counts s^{-1} is expected to be at least $\sim\sqrt{30} \approx 6$ counts s^{-1}. This peak lies on a background of 50 ± 10 counts s^{-1}, so the overall standard deviation in the net signal is $(10^2 + 6^2)^{1/2} \approx 12$ counts s^{-1}, i.e., 40% of the net signal.

The linear dynamic range for a given element in ICP–MS goes from the minimum quantifiable concentration to the analyte concentration that yields on the order of 10^6 counts s^{-1}. At higher count rates, pulse pileup occurs in the multiplier and counting electronics; the former is probably the chief culprit. This linear range is comparable to that observed in ICP–AES (*1–3*) and is certainly adequate for many analytical purposes. However, a detector with a higher maximum count rate capability is still desirable for isotope ratio measurements with rare isotopes or simultaneous measurement of trace and major constituents. The Channeltron detector also has a finite useful lifetime of about one year, after which it loses gain and needs to be replaced.

The spectra of Figure 5 illustrate that ICP–MS can also provide isotope ratio information. For this type of application the precision achievable and the amount of analyte required are primary figures of merit. In general, ICP–MS can determine isotope ratios with a precision of 0.1–3% relative standard deviation (RSD), with poorer precision generally obtained for very large or small ratios or as the element's concentration decreases. As a rule of thumb, 2–10 mL of solution containing the analyte at a concentration of at least 10 $\mu g\ L^{-1}$ is required, corresponding to nanogram amounts of analyte. This figure is for sample introduction by conventional solution nebulization and likely can be improved considerably by use of more efficient methods such as a recirculating nebulizer or volatile compound gener-

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ation, e.g., OsO_4 for Os isotope ratio determinations (*36*).

The benchmark technique for isotope ratio determination, namely thermal ionization MS, is indeed capable of better precision (0.01% RSD) with less analyte. However, ICP–MS has some additional attractive features in that isotope ratios can be determined on a multielement basis provided, of course, that the various isotopes of interest do not overlap each other or background peaks. In cases where separation of the element(s) of interest is not necessary, ICP–MS is potentially very rapid and has attracted attention for applications requiring isotope ratio determinations on large numbers of samples, e.g., geochemical prospecting using Pb isotope ratios. In such an application the concentrations of various trace elements can also be determined and correlated with the isotope ratios for the target element(s). Isotope ratio determinations for some elements in some matrices will require prior chemical separations. A veritable arsenal of such separation procedures has been developed, which will be as useful with ICP–MS as with other measurement methods. Finally, the isotopic capability of ICP–MS makes stable isotope dilution a valuable tool for quantitative analysis of those elements with more than one stable isotope.

The final, and perhaps most important, figures of merit to be considered are accuracy, precision, and susceptibility to interference. To this end, Date and co-workers at the British Geological Survey conducted comprehensive elemental analyses of virtually all the available geological standard reference materials using ICP–MS. In general, the agreement between the determined concentrations and the certified values has been very good, i.e., ± 10% or better, even for elements at sub-$\mu g\ g^{-1}$ levels in the original sample. Many of the materials chosen contained high levels of Fe, rare earths, or other elements with overly rich emission spectra, which would have caused spectral overlap problems in ICP–AES (*35*, *37*).

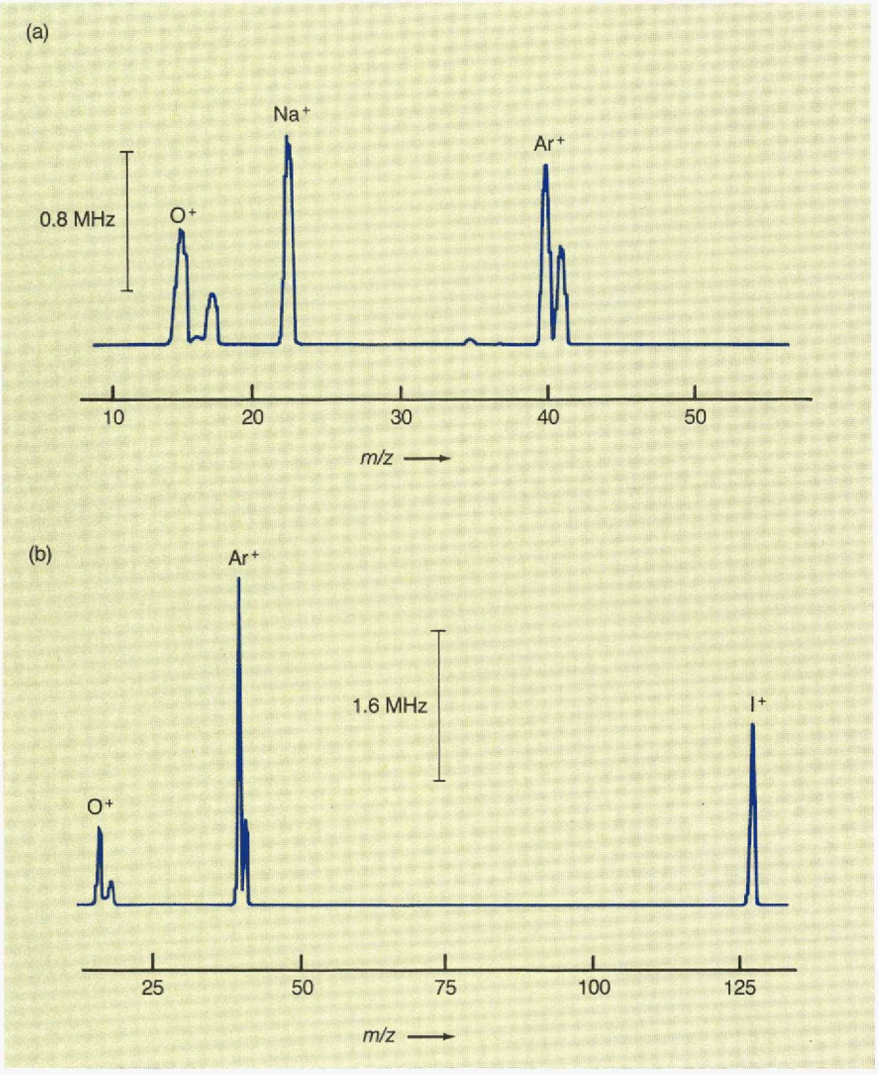


Figure 6. Mass spectra depicting contribution of matrix ions to total ions observed
(a) 0.020 M NaCl and (b) 0.020 M NH_4I. Analog ion current detection with current-to-frequency conversion. Reproduced from Reference 31

The first wave of commercial ICP–MS instruments is now in the analytical community, and the users of these instruments are the ones evaluating the critical figures of merit mentioned in the preceding paragraph. In principle, short-term precision and accuracy should be on the order of that obtainable by ICP–AES, namely a few percent relative. However, various users of the first commercial instruments have reported difficulties with drift of sensitivity with time. For example, our experience in Ames with our Sciex instrument has been that the slope and/or the intercept of the calibration curve tend to drift down at a rate of 5–10% per hour. Several potential sources of drift have been identified; these include temperature-dependent fluctuations in the quadrupole power supply (*38*) and gradual solid deposition on the orifice. The manufacturers claim to have eliminated the first problem. The second phenomenon is minimized by keeping the total solute levels below approximately 0.2% if the solution is to be nebulized continuously for any length of time. Sciex now claims to have developed a refined ion optical system that reduces the rate of signal drift to 1–4% per hour and improves reproducibility to ±1% over 6 h if internal standardization is used (*39*). Continued work on further improving long-term stability is no doubt progressing at both Sciex and VG.

The recommended upper limit on total solute level has another practical cause in that ICP–MS seems to be somewhat more susceptible to ionization suppression effects than ICP–AES (*31*). The basis for ionization suppression stems from the inherent efficiency of the ICP as a universal ion source for nearly all the elements in a sample. The effect is discernible from the spectra shown in Figures 4 and 6. The blank spectrum (Figure 4) represents the ambient ion composition of the ICP modified by whatever reactions occur during ion extraction. Because the ICP is electrically neutral and doubly charged ions are few, the total ion density seen by the MS is related to the total electron density in the region sampled. Ultrasonic nebulization of 0.020 M NaCl (0.12%) yields a Na^+ peak that is approximately 30% of the total ion signal observed (Figure 6). Thus the total electron density in the ICP depends on the amount of Na present in the sample solution.

Changes in Na concentration from sample to sample will affect the analyte ion–atom ratio in the ICP by changing the electron density in Equation 1.

$$M \rightleftarrows M^+ + e^- \qquad (1)$$

For example, the Co^+ signal observed in the presence of 0.02 M (0.12%) NaCl was approximately 70% of that in the absence of Na (*36*). When a similar ultrasonic nebulization facility was used in ICP–AES, some 10 times as much NaCl was required to suppress ionic emission by the same extent (*33*). A similar disparity in the extent of ionization suppression is also found between ICP–MS and ICP–AES when a conventional pneumatic nebulizer is used.

In ICP–MS even matrix elements such as I (ionization energy = 10.45 eV), which are not normally considered easily ionized, can suppress analyte ionization if present in sufficient excess (*31*). The reasons that ionization suppression should occur at lower solute levels in ICP–MS relative to ICP–AES are not well understood at this time but may be related to processes occurring in the ion extraction process in ICP–MS. As was the case for spectral overlap from oxide or doubly charged ions, ionization suppression is readily detectable by scanning the mass spectrum and verifying that the ratio of matrix ions to total ions is either small or reproducible from sample to sample. In addition, some degree of ionization suppression is definitely tolerable if isotope dilution is used, and preliminary evidence indicates that internal standardization will correct for ionization suppression to some extent.

Conclusion

The alliance of the ICP with MS shows promise of extending the sensitivity and selectivity of MS to elemental analysis of solutions. The next few months will see the crucial studies of the extent to which the excellent powers of detection of ICP–MS will really prove useful in actual analytical problems.

Some additional areas of research interest for ICP–MS include the following:

- its use as an element and isotope selective detector for chromatography, flow injection analysis, laser ablation (*40*), and other sample introduction techniques,
- fundamental studies of ionization and other basic processes in the ICP (*41*, *42*),
- detection of halogens and perhaps other elements as negative ions, and
- auxiliary ionization of neutral analyte atoms formed from ICP ions that have recombined with electrons during the extraction process.

The latter concept is particularly interesting because the present detection limits for ICP–MS are achieved even though only one analyte ion is detected for every 10^5 or 10^6 that would be going through the skimmer if no recombination occurred (*21*)! ICP–MS with a different type of mass analyzer that can resolve isobaric ions and/or detect ions at more than one mass simultaneously may also prove valuable. The present work merely represents an interim report on a technique that still has considerable room for improvement in performance and basic understanding.

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